

COMPLETING THE SQUARE

1. Using cardboard or poster board, construct a square.
 - a. The square should be large enough for a presentation.
 - b. Label each side x .
 - c. Write x^2 in the center. This is the area of the square.
2. Construct a rectangle with length x (same as the square from step 1) and width b .
 - a. b can be any width you like, but it would be best if b was between a third to one-half of x .
 - b. Write bx in the center of the rectangle. This is the area of the rectangle.
3. Explain how the square and rectangle represent $x^2 + bx$.
4. Make an exact copy of the rectangle " bx " (from step 2).
 - a. Cut this rectangle in half.
 - b. Label the sides of both of these rectangles with $b/2$ and x appropriately.
5. Arrange the " x^2 " square and the two " $b/2$ " rectangles as follows:
 - a. Place one of the " $b/2$ " rectangles so that its side of length x is paired with a side of the " x^2 " square (this will form a rectangle).
 - b. Next place the other " $b/2$ " rectangle so that it is orthogonal to the other " $b/2$ " rectangle and also paired with a side of the " x^2 " square (this will look like a big square with a little square corner missing).
6. Make a little square that will fit into the missing corner in the above step.
 - a. Place this little square into the above step to "complete the square". The result should be a big square.
 - b. This little square has an area of $(b/2)^2$; explain.
7. Determine the area of the big square from the above step.
8. Explain how the steps above relate to the equation: $x^2 + bx + (b/2)^2 = (x + b/2)^2$.
9. Relate how the concepts in the above steps can be used to write $x^2 + 6x + 7$ in the form $(x + 3)^2 - 2$.
10. Write an essay or prepare a presentation explaining the steps above. Show all your work and include the shapes from steps 1-4.